



FOUR-COURSE SYSTEM

COURSE 3 - DATA ENGINEERING 2026

C3-041 Practice Pack

Attempt-protected independent tasks. Use the starter ZIP; do not open matching review before saving a genuine attempt.

Artifact	C3-041
Status	2026 - LEARNER EDITION
Phase	C3-03 Orchestration & Reliable Workflows 13 lessons 34-46 focused hours
Prerequisite	C3-02 Gate PASS; C2B Python/quality/ownership foundations assumed, orchestration taught here.
Brand	Charcoal #1F2937 Teal #14B8A6 Soft Blue #3B82F6 AMOLED #000

House teaching rule: simple mental picture -> tiny example -> guided Airflow/tool workflow -> break/fix -> independent changed task -> validation -> production reasoning -> evidence. A green scheduler box alone never proves correct data.



Practice rules

Every Pxx is an independent task after the guided lesson. Save code/config/output, one validation control, confidence 1-4 and the main uncertainty. Attempt history is append-only.



P01 - Workflow orchestration and DAG design

Independent task

Create `architecture/DAG\_DESIGN.md` for the supplied practice pipeline. State each task's input, output, side effect, retry boundary and downstream dependency.

Change it so this is not copied practice

Redesign the same workflow when extraction comes from an API but loading still targets the same warehouse. Defend which task boundaries change and which do not.

Your evidence must include

Saved DAG design + dependency reasoning + one failure-path walkthrough + changed transfer.

Independent validation

Every task has a single operational responsibility; publish is blocked by validation/reconciliation; no task depends on hidden local leftovers.

Failure/recovery note

Record one failure, surprise or uncertainty. State what evidence you inspected before changing code/config. If no failure occurred, inject the lesson's controlled break/fix case and recover it.

Attempt log - append only

Attempt ID/date: _____ | confidence 1-4: ____ | main uncertainty: _____ | independent control: _____

Do not open R01 until this attempt is saved.



P02 - Airflow setup, standalone mode and project structure

Independent task

Follow `AIRFLOW\_SETUP\_CHECKLIST.md`; save command/version evidence and a project-tree snapshot. Do not paste generated admin credentials into evidence files.

Change it so this is not copied practice

Delete only the disposable venv/runtime state, recreate the environment, and prove the project source still parses/discovers.

Your evidence must include

Version/startup/discovery evidence + clean project tree + secret hygiene check + recreated-environment proof.

Independent validation

Airflow reports 3.3.1; standalone starts; source project is separate from metadata/logs/secrets; no credential appears in Git-tracked files.

Failure/recovery note

Record one failure, surprise or uncertainty. State what evidence you inspected before changing code/config. If no failure occurred, inject the lesson's controlled break/fix case and recover it.

Attempt log - append only

Attempt ID/date: _____ | confidence 1-4: ____ | main uncertainty: _____ | independent control: _____

Do not open R02 until this attempt is saved.



P03 - DAG authoring with TaskFlow and tasks

Independent task

Complete the TODO TaskFlow DAG skeleton without copying business logic into the DAG file. Keep DAG/task IDs stable and make dependencies visible.

Change it so this is not copied practice

Add a second compact metadata output (for example `row_count`) without passing the underlying dataset through XCom.

Your evidence must include

Runnable DAG + graph evidence + pure-function test evidence + XCom/data-boundary explanation.

Independent validation

DAG imports from ``airflow.sdk``; import time has no I/O side effects; business logic remains testable outside Airflow; dependencies match the intended graph.

Failure/recovery note

Record one failure, surprise or uncertainty. State what evidence you inspected before changing code/config. If no failure occurred, inject the lesson's controlled break/fix case and recover it.

Attempt log - append only

Attempt ID/date: _____ | confidence 1-4: ____ | main uncertainty: _____ | independent control:

Do not open R03 until this attempt is saved.



P04 - Schedules, logical dates and data intervals

Independent task

Implement interval-derived input/output paths in the practice DAG and save evidence for two different logical dates.

Change it so this is not copied practice

Change the schedule from daily to a different interval and restate what one run represents before editing path logic.

Your evidence must include

Two interval proofs + derived paths + catchup rationale + changed-schedule transfer.

Independent validation

Historical run always maps to historical input/output; two intervals cannot overwrite each other; start date is timezone-aware; catchup choice is documented.

Failure/recovery note

Record one failure, surprise or uncertainty. State what evidence you inspected before changing code/config. If no failure occurred, inject the lesson's controlled break/fix case and recover it.

Attempt log - append only

Attempt ID/date: _____ | confidence 1-4: ____ | main uncertainty: _____ | independent control:

Do not open R04 until this attempt is saved.



P05 - Parameters, variables, connections and secret boundaries

Independent task

Complete `CONFIG\_SECRET\_BOUNDARY.md`; configure the practice connection/pool/parameter using non-secret examples and prove a repository scan contains no credential literals.

Change it so this is not copied practice

Move the same DAG from local to another environment without changing DAG code; list which environment-owned values must be supplied.

Your evidence must include

Boundary table + safe config example + secret scan + changed-environment transfer.

Independent validation

No secret literal in tracked source; per-run values are explicit; connection identifiers are stable; config ownership/precedence is documented.

Failure/recovery note

Record one failure, surprise or uncertainty. State what evidence you inspected before changing code/config. If no failure occurred, inject the lesson's controlled break/fix case and recover it.

Attempt log - append only

Attempt ID/date: _____ | confidence 1-4: ____ | main uncertainty: _____ | independent control: _____

Do not open R05 until this attempt is saved.



P06 - Retries, timeouts and failure callbacks

Independent task

Configure bounded retries/timeouts on the designated task, trigger the fail-once path, and save first-failure + retry-success evidence. Then prove deterministic bad data remains a clear failure.

Change it so this is not copied practice

Change the dependency from local file to rate-limited API and redesign retry/backoff without changing the business validation policy.

Your evidence must include

Policy table + fail-once evidence + permanent-failure evidence + timeout rationale.

Independent validation

Transient path shows failed attempt then success; permanent bad-data path stays failed; side effects remain idempotent; failure log retains run/task/partition context.

Failure/recovery note

Record one failure, surprise or uncertainty. State what evidence you inspected before changing code/config. If no failure occurred, inject the lesson's controlled break/fix case and recover it.

Attempt log - append only

Attempt ID/date: _____ | confidence 1-4: ____ | main uncertainty: _____ | independent control: _____

Do not open R06 until this attempt is saved.



P07 - Catchup, backfills and controlled reprocessing

Independent task

Use the backfill workspace to plan/dry-run and then execute the permitted historical range in the learner Airflow runtime. Save run range, behavior, concurrency and reconciliation evidence.

Change it so this is not copied practice

Repeat with `failed`-only reprocessing and explain why it differs from deliberately recreating completed runs.

Your evidence must include

Backfill plan + run evidence + stale-safe proof + rerun proof + reconciliation.

Independent validation

Correct historical partitions; no duplicate current keys; newest valid version retained; rerun materially unchanged; final totals reconcile.

Failure/recovery note

Record one failure, surprise or uncertainty. State what evidence you inspected before changing code/config. If no failure occurred, inject the lesson's controlled break/fix case and recover it.

Attempt log - append only

Attempt ID/date: _____ | confidence 1-4: ____ | main uncertainty: _____ | independent control:

Do not open R07 until this attempt is saved.



P08 - Sensors and data-aware dependencies

Independent task

Add/configure the input readiness wait in the practice DAG; test both present-file and missing-file behavior in the learner runtime.

Change it so this is not copied practice

Replace filesystem readiness with an upstream marker/table condition and preserve the same readiness/timeout contract.

Your evidence must include

Readiness contract + success evidence + timeout evidence + changed-dependency transfer.

Independent validation

Downstream is blocked before readiness; present input proceeds; missing input fails within bound; waiting does not use a custom busy loop.

Failure/recovery note

Record one failure, surprise or uncertainty. State what evidence you inspected before changing code/config. If no failure occurred, inject the lesson's controlled break/fix case and recover it.

Attempt log - append only

Attempt ID/date: _____ | confidence 1-4: ____ | main uncertainty: _____ | independent control:

Do not open R08 until this attempt is saved.



P09 - Idempotent task design and partial reruns

Independent task

Complete the TODO idempotence/partial-rerun notes; run the pure pipeline twice for one partition and save before/after key/count/amount controls.

Change it so this is not copied practice

Change one source row to a stale earlier version and prove replay does not regress the current record.

Your evidence must include

Two-run reconciliation + stale-version test + partial-rerun explanation.

Independent validation

Same logical input twice gives same business state; no duplicate keys; current version rule preserved; partial rerun touches only intended outputs.

Failure/recovery note

Record one failure, surprise or uncertainty. State what evidence you inspected before changing code/config. If no failure occurred, inject the lesson's controlled break/fix case and recover it.

Attempt log - append only

Attempt ID/date: _____ | confidence 1-4: ____ | main uncertainty: _____ | independent control:

Do not open R09 until this attempt is saved.



P10 - Pools, concurrency and resource limits

Independent task

Create/configure the practice pool, attach the designated tasks, document the protected resource and save queue/running evidence in the learner Airflow runtime.

Change it so this is not copied practice

Move the same workflow to a slower API and redesign the pool/concurrency limits from the dependency's contract.

Your evidence must include

Pool config + resource rationale + observed scheduling evidence + changed-resource transfer.

Independent validation

Tasks consuming the scarce resource share an explicit pool; limits are justified; a backfill cannot launch unbounded concurrent writes.

Failure/recovery note

Record one failure, surprise or uncertainty. State what evidence you inspected before changing code/config. If no failure occurred, inject the lesson's controlled break/fix case and recover it.

Attempt log - append only

Attempt ID/date: _____ | confidence 1-4: ____ | main uncertainty: _____ | independent control: _____

Do not open R10 until this attempt is saved.



P11 - DAG testing and pre-deployment validation

Independent task

Build `PRE\_DEPLOY\_CHECKLIST.md` and make the supplied checks pass without deleting assertions or hiding import errors.

Change it so this is not copied practice

Add a fresh failing fixture to a pure function test and show that pre-deploy QA blocks the change before the DAG is scheduled.

Your evidence must include

Checklist + passing tests/static checks + learner Airflow discovery/test evidence + changed failing fixture proof.

Independent validation

Pure tests pass; DAG compiles/imports; no import-time external work; dependency graph is intentional; runtime checks are clearly distinguished from static checks.

Failure/recovery note

Record one failure, surprise or uncertainty. State what evidence you inspected before changing code/config. If no failure occurred, inject the lesson's controlled break/fix case and recover it.

Attempt log - append only

Attempt ID/date: _____ | confidence 1-4: ____ | main uncertainty: _____ | independent control: _____

Do not open R11 until this attempt is saved.



P12 - Operational logs, metadata and troubleshooting

Independent task

Complete `TROUBLESHOOTING\_WORKSHEET.md` from actual practice evidence; every diagnosis must name DAG/run/task/interval/symptom/evidence/recovery.

Change it so this is not copied practice

Diagnose a fresh 'queued but not running' case caused by a deliberately constrained pool rather than code failure.

Your evidence must include

Four incident worksheets + one fresh constrained-pool diagnosis + recovery/reconciliation evidence.

Independent validation

Diagnosis names the failure layer and evidence; recovery is minimal and idempotent; no secret values are copied into evidence.

Failure/recovery note

Record one failure, surprise or uncertainty. State what evidence you inspected before changing code/config. If no failure occurred, inject the lesson's controlled break/fix case and recover it.

Attempt log - append only

Attempt ID/date: _____ | confidence 1-4: ____ | main uncertainty: _____ | independent control: _____

Do not open R12 until this attempt is saved.



P13 - Deployment patterns, recovery runbooks and handoff

Independent task

Complete `runbooks/RUNBOOK\_TODO.md` and `DEPLOY\_HANDOFF.md`; have a second-person simulation checklist explain how to recover the missing-input and partial-publish incidents.

Change it so this is not copied practice

Assume the next release changes one task ID and one connection requirement. Write the migration/rollback note that prevents accidental historical confusion.

Your evidence must include

README + deploy handoff + runbook + rollback/reprocess path + second-person simulation evidence.

Independent validation

No secrets/runtime state bundled; version/source identity recorded; deployment and recovery are reproducible; runbook includes validation after recovery.

Failure/recovery note

Record one failure, surprise or uncertainty. State what evidence you inspected before changing code/config. If no failure occurred, inject the lesson's controlled break/fix case and recover it.

Attempt log - append only

Attempt ID/date: _____ | confidence 1-4: ____ | main uncertainty: _____ | independent control:

Do not open R13 until this attempt is saved.